

J2EE

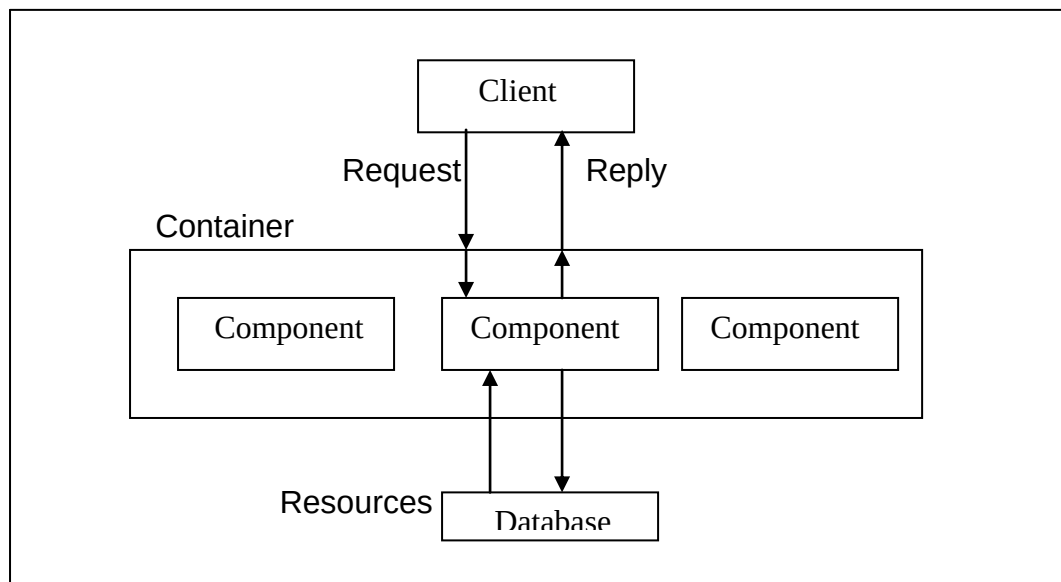
1. Explain about Multi-tier architecture

A tier is an abstract concept that defines a group of technologies that provides one or more services to its clients. In multi-tier architecture each tier contains services that include software object or DBMS. Multi-tier architecture is composed of clients, resources, components (service), and containers.

Clients, Resources and Components

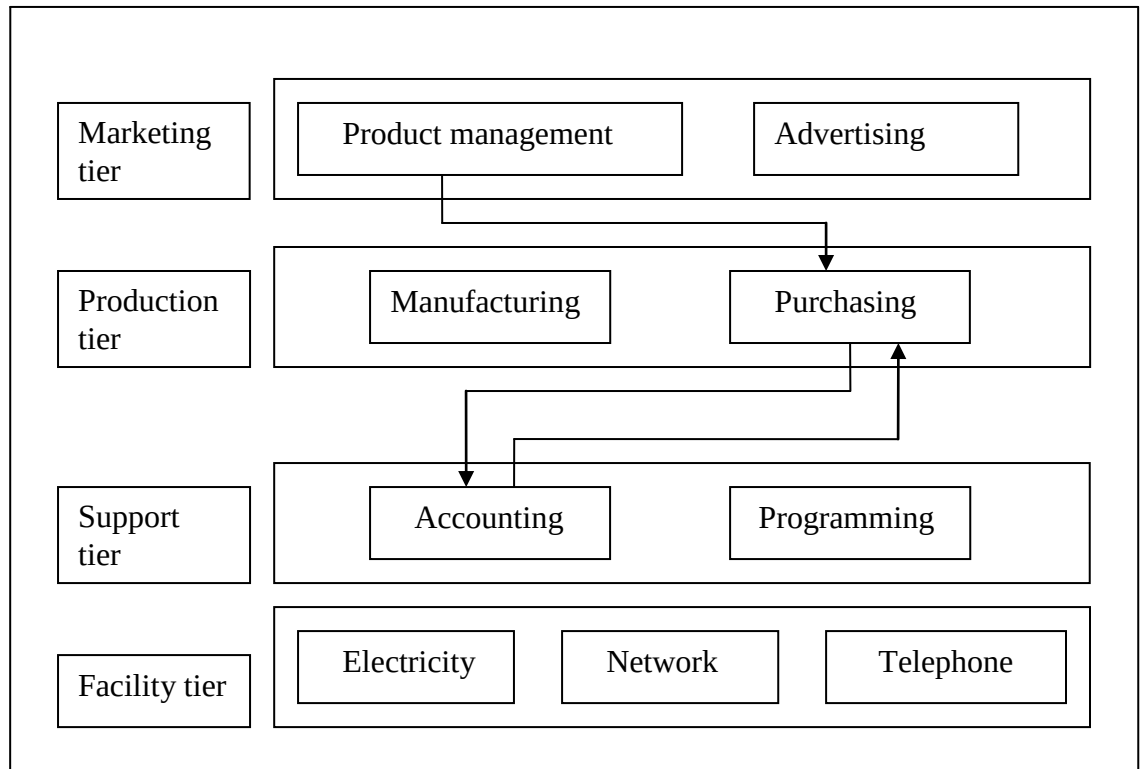
A client refers to a program that requests a service from a component. A resource is anything a component needs to provide a service, and a component is part of tier that consists of a collection of classes or a program that performs a function to provide the service. A container is software that manages a component and provides a component with system services. The relationship between a container and a component is sometimes referred to as a contract, whose terms are governed by an application programming interface (API). An API defines rules a component must follow and the services a component will receive from the container.

A container handles persistence, resource management, security, threading and other system-level services for components that are associated with the container. Components are responsible for implementation of business logic. It helps the programmer to focus on coding business rules into components without becoming concerned about low-level system services. The relationship between client, component and resource is shown below.



Normally large organizations employ multi-tier architecture because it is easy to build an application that is flexible, scalable and responsive to the expectation of clients. Considering an organization that groups its services as marketing tier, production tier, support tier and facility services tier. At the lowest level facility services contains variety of resources that include

electricity, elevator, computer network, and telephone services. The next tier in the organization contains support resources like computer programming, accounting, counseling etc. Production tier has the resources necessary to produce products and services sold by the company. The highest tier consists of resources for product management and advertising. All the tiers should interact with each other for the proper functioning of the enterprise. This is similar to the tier structure in distributed systems. The tier relationship in an enterprise is given below.



2. Explain in detail about J2EE architecture

J2EE is four-tier architecture. These consist of Client Tier (Presentation tier or Application tier), Web tier, Enterprise JavaBeans Tier(or Business tier), and the Enterprise Information Systems Tier. Two or more tiers can physically reside on the same Java Virtual Machine although each tier provides a specific type of functionality to an application. Some of the APIs of J2EE components can be used on more than one tier (i.e. XML API), while other APIs (i.e., EJB API) or associated with a particular tier.

CLIENT TIER

Client tier consists of programs that interact with the user. It prompts the user for input and then convert the user's response into requests that are forwarded to software on a component that processes the request and returns results to the client program. J2EE clients can be classified as follows

Web client is a A software(usually browser) that accesses resources located on the web tier.

Ejb client can access one or more enterprise javabeans that are located on the EJB tier rather than resources on the web tier.

EIS clients are the interface between users and resources located on the EIS tier.

Multi-tier clients can access components located on tiers other than the tier where the multi-tier client resides.

WEB TIER

Web tier accepts requests from other software that was sent using POST, GET, and PUT operations, which are part of HTTP transmissions. The two major components of web tier are Servlets and Java Server Pages. A servlet is a java class that resides on the web tier and is called by a request from a browser client that operates on the client tier. A servlet is associated with a URL that is mapped by the servlet container. It typically generates an HTML output stream that is returned to the web server. The web server in turn transmits the data to the client. JSP is different than a servlet depending on the container that is used. JSP uses custom tags to access the bean.

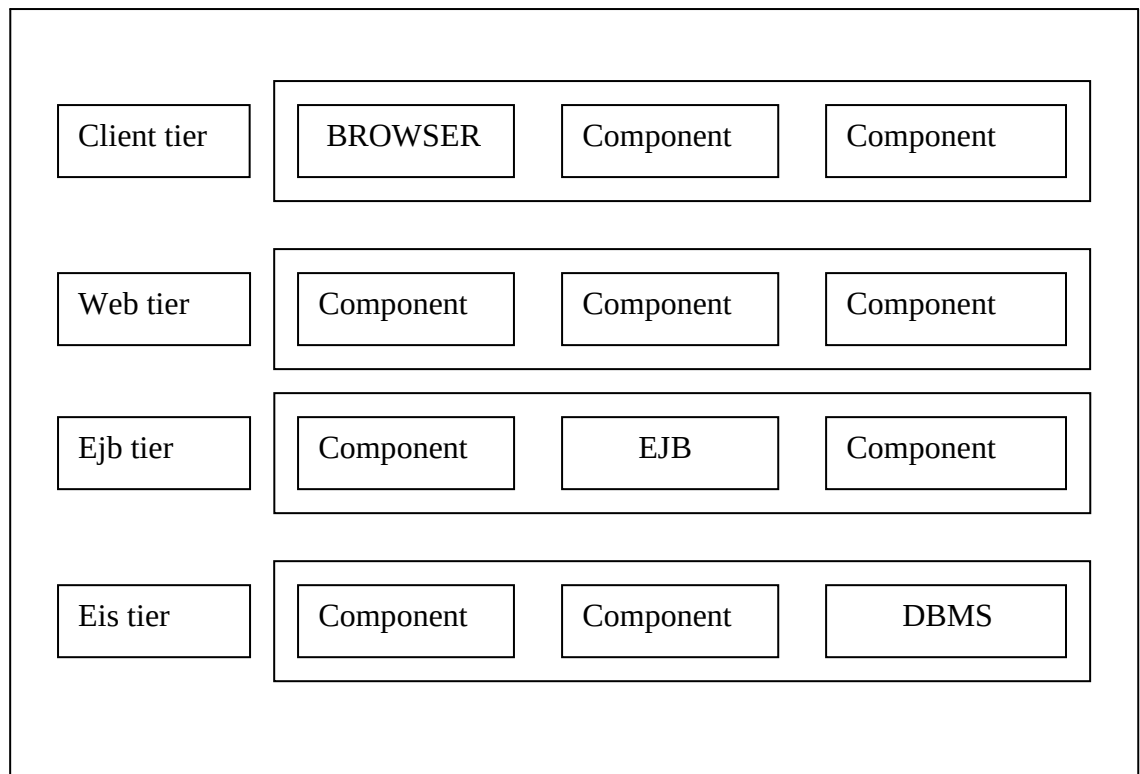
ENTERPRISE JAVA BEANS TIER

Enterprise java bean is a class that contains business logic and callable from a servlet or Jsp. EJB tier contains the enterprise java beans server that stores and manages enterprise java beans. This tier automatically handles concurrency issues that assure multiple clients have simultaneous access to the same object and also manages instances of components. EJB server and EJB container is responsible for low level system services that are essential for implementing business logic.

ENTERPRISE INFORMATION SYSTEMS TIER

This tier provides flexibility to developers of J2EE applications since it include variety of resources and support connectivity to resources. It defines all the elements that are needed to communicate between J2EE application and non-J2EE software.

The following figure shows the tiers of J2EE that provides a specific function to an application.



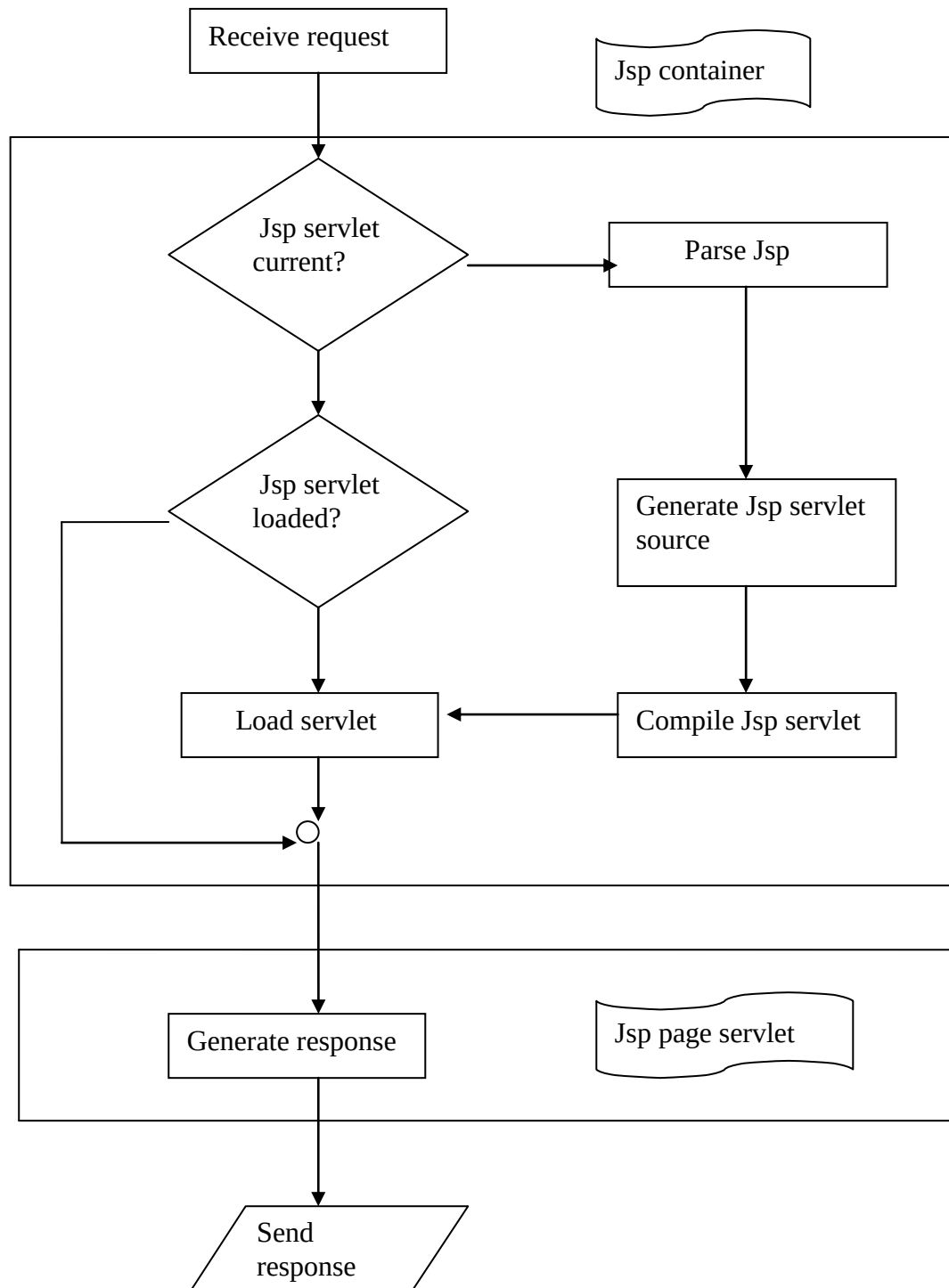
3. How are Java server Pages executed

A JSP execution start with a request for a JSP page and the processing is performed based on the JSP tags present in the page in order to generate content dynamically. The web containers call the Servlets for all requests with URL that match the jsp extension. Servlets not only finds the jsp pages in response to each request but also compiles them. Each jsp page is compiled into a page specific servlet that generates dynamic content specified by the original jsp document.

Initially, the jsp page compiler parses through its contents, looking for jsp tags. As it parses the file, it translates its contents into the equivalent java source code. When executed it will generate the output indicated by the contents of the original file. When the servlet code is constructed, the page compiler servlet calls the java compiler to compile the source code and add the resulting java class file to the relevant directory in the jsp container classpath. When the compiled jsp page servlet is ready, the page compiler servlet invokes the new servlet to generate the response for the original request. These steps are required only for the first time a jsp page is requested. For all the subsequent requests, it is parsed directly to the already-compiled page servlet.

When a jsp page compiler receives a jsp request, it checks the time stamp of the jsp file to determine whether the file is modified or created. It also checks the time stamp on the compiled servlet for the javaserverpage. If the time stamp on the jsp file is more recent than the one on the compiled page

servlet, then a new servlet is generated. Server process for creating and running jsp servlets is given below.



4. Explain the tags used in jsp? How does a jsp refer the bean?

Comment tag

Three types of comments are available in JSP comment, content comment and scripting language comment.

Content comment

```
<!-- <%= fact(20%)> -->
```

It returns the calculated values as part of the response from browser.

JSP comments

It is independent of scripting language used by the page. It can be viewed by

renaming the original JSP file.

```
<% - - comment -- %>
```

Scripting language comment

It has native comment syntax of the scripting language

```
<% /* comment */ %>
```

Declaration tag

```
<% ! int i=5 %>
```

```
<% ! public long fact()
```

```
{
```

```
}
```

```
%>
```

Expression tag

```
<% = expression %>
```

Directive tag

It opens with <% @ and commands the JSP virtual engine to perform a specific task and close with %>. The commonly used directives are

import - To import java packages into JSP program

include - Insert a specified file into JSP program

taglib - Specifies a file that contains a tag library

```
<% page import = "package name" %>
```

```
<% include file = "file name" %>
```

```
<% taglib uri = "libraryfile.tld" %>
```

Scriptlet tags

It contains commonly used java control statements and loops within `<% -- %>`

JSP can access the bean using three tags namely use Bean, get property and set property tags. These tags allow the JSP to place beans into the page as well as alter and access their properties.

- **Use Bean**

```
<jsp: use Bean  id="instance name"  class="Name of the class file"
               scope ="session/page/application">
```

The tag is used to create a Bean instance or fetch existing one from the server. The id is used to refer a particular bean throughout the page. More than one bean can be used within a single page.

- **Get property**

It access the beans properties in JSP and produces the content that we can see in the HTML generated by the page. The property name should match with the methods defined in the bean.

```
<jsp: get property name= "Bean instance name"  property = "property
name">
```

- **Set property**

This tag is used anywhere within the page to modify the bean properties.

```
< jsp: setproperty name= "Bea instance"
                  property = "property name"
                  value   = "Direct value/ expression" %>
```

5. What is enterprise java bean? Explain the steps to create stateless session bean?

An EJB is a server side component of the J2EE architecture that provides business logic to a j2ee application and interacts with other server side j2ee components. An ejb is written in java, so it is platform and operating system independent. An ejb is comprised in an ejb container which is present inside the ejb server. Enterprise beans are classified into two types namely session bean and entity bean. Session bean persists only for a session and they cannot update the database but an entity bean can access and update the database. Session bean is classified into two types namely stateless session bean and state full session bean. Stateless session bean does not maintain

conversation between two states, but a state full session bean can maintain conversations.

An ejb program has remote interface, Home interface and ejbbeanclass in the server side. The client program can call the business method in the bean class using home and remote interfaces. JNDI is used to link the server program and client program.

Remote Interface

It defines the business methods that a client may call. These methods are implemented in the enterprise bean code. Every remote interface must extend javax.ejb.EJBObject interface and each method must throw remote exception.

Eg. Public interface conversion extends EJBObject

```
{  
    public double RupeestoDollar(double rup) throw RemoteException  
}
```

Home Interface

This interface defines the methods that allow a client to create, find, or remove an enterprise bean. It should have a create method, which initialize an instance of an bean and returns an object that is an instance of remote interface.

Eg. Public interface conversion Home extends EJBHome

```
{  
    Conversion create() throws Remote Exception, Create Exception  
}
```

Enterprise Bean class

This bean implements the actual business method that a remote interface defines. The session bean should implement javax.ejb.sessionbean interface. When the container parses the sessioncontext interface, a sessioncontext is associated with the instance for the lifetime of the bean.

E.g. Public class conversionEJB implements SessionBean

```
{  
    public double RupeestoDollar(double rupees) {
```



```

        -----steps-----    }

    public void ejbCreat()

    public void ejbremove()

    public void ejbActivate()

    public void ejbpassivate()

    public void setSessionContext()

}

```

6. What is entity bean? Describe the life cycle of entity bean

An entity bean represents an entity kept in a persistent storage mechanism, normally a database. Entity beans are persistent, allow shared access and have primary keys.

Features

- Can be used concurrently by several clients
- Can survive even after the server is crashed
- They are long-lived. They exist even after the lifetime of a single client

LIFE CYCLE

An entity bean can be in one of the following states among Non-existence, Pooled, and Ready.

Does not exist: Initially a bean is in non-existence state when it is not used. When the new Instance () and setEntityContext () methods are invoked on a bean, it enters the pooled from non-existence state. When finalize () method is called it moves from pooled state to non-existence state.

Pooled state: When ejbFind () method is invoked, the container uses an EJB instance from the pooled state to handle the request. The bean is moved from pooled state to ready state when the container selects the instance to handle an ejbCreate () request, at which point create and post create are invoked on

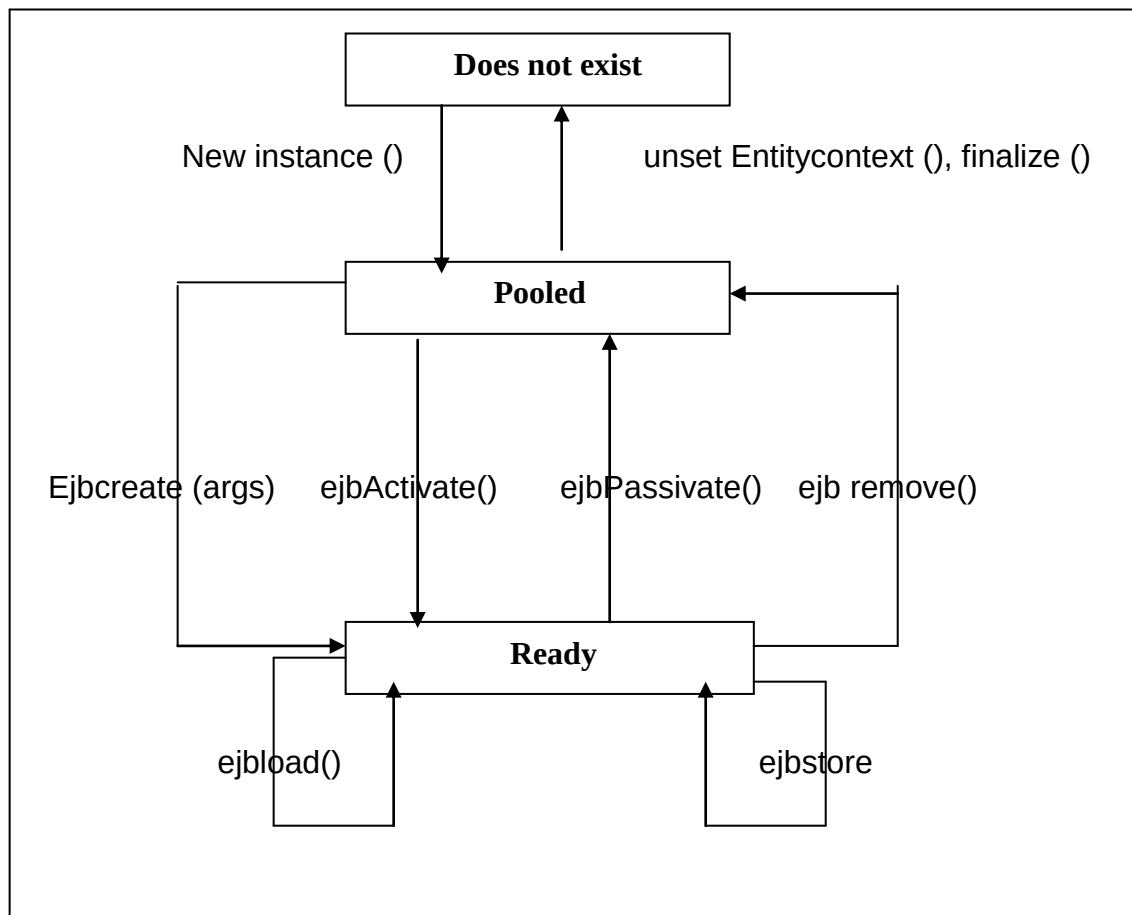
the object. The container can also select the instance to be activated at which point `ejbActivate()` is invoked on the instance.

When the container selects the instance for passivation, at which point `ejbPassivate()` is called on the object so that it is moved from ready state to the pooled state. When a bean returns to the pooled state, it once again loses its identity and becomes one of the several available instances in the pool.

Ready State

When the bean is in the ready state, it is able to accept invocations of its business methods. It can also process its `ejbload()` and `ejbstore()` methods. The container invokes these methods to keep the state of the object consistent with its representation in the database. The `ejbstore()` method writes data from the instance variables to the database.

LIFECYCLE OF ENTITY BEAN



7. Write a Bean program to encapsulate shopping cart problem. Write a jsp that displays the contents of the shopping cart and uses the created shopping cart.
8. Write a jsp to display a set of quiz questions and accept the answers from users. Write a jsp to evaluate the questions and print the score. Write a timer bean program for counting the lapsed time for answering the quiz.
9. Write a bean program to establish a set of connections from database and initialize the connection. Write a jsp to use the pooled connection to access different tables from a database simultaneously.
10. Write a java bean program using database to post and update current news. Write a jsp to view the news based on selected topics.